

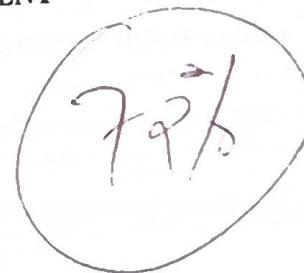
THE ABERDARE TECHNICAL COLLEGE
ICT TECHNICIAN LEVEL 5

UNIT OF COMPETENCE: NETWORK DESIGN AND MANAGEMENT

UNIT CODE: IT/CU/ICTA/CR/02/5/MA

CONTINUOUS ASSESSMENT TEST (CAT) 1, TERM 3

TVET CDACC ICT LEVEL 5



Time: 2 Hours

Marks: 50 Marks

NAME: YAHYA MARCHIS DATE: 25/5/20

CDACC REGNO.....

INSTRUCTIONS

- ✓ Answer all questions.
- ✓ Write your answers clearly.
- ✓ Show all workings where necessary.
- ✓ Write your students information on the slots provided
- ✓ The test consists THREE pages and NINE questions.
- ✓ DO NOT remove any page from this test

1. Define the following terms: (4 Marks)

- a) Half-duplex
- b) Node
- c) Bandwidth
- d) Topology

2. Differentiate between LAN and WAN. (4 Marks)

3. State FOUR advantages of star topology. (4 Marks)

4. Explain the following network topologies: (10 Marks)

- a) Star topology
- b) Ring topology
- c) partial mesh topology
- d) Hybrid topology

1. Half duplex - This is data transmission where only one device can transmit data at any given moment. ✓

Node -> Any device connected to a network that can send, receive or forward data.

Bandwidth -> This maximum capacity of data transmitted at a given period. ✓

Topology -> Physical or logical arrangement of devices in a network. ✓

2. Difference of LAN and WAN

Local area network is used to connect devices in a small geographical area like in single building while ~~wide~~ ^{WAN} area network connects devices or LAN across large geographical area like cities. ✓

3. Advantage of star topology

Has High reliability. ✓

Easy to maintain. ✓

Better Performance and high speed. ✓

4. Explain ~~various~~ topologies

Star topology.

This is a topology where all devices are connected to a central device either switch or hub. ✓

Ring topology.

All devices are connected into a closed loop or circle and data travels in one direction of the ring. ✓

Partial mesh topology

This topology some devices are connected to many other devices but not all are connected to many devices like full mesh topology. ✓

Hybrid topology.

This topology include combination of two or more topologies like ring and mesh. ✓

Point to point topology.

This topology include two devices that are connected to each other directly using single communication links or cable. ✓

5 Components of Computer network.

Host or end devices: like Computer. ✓

Transmission Media like fibre optic cable. ✓

Network devices like Switches. ✓

Network protocols like TCP/IP. ✓

OF

6 Function of router in computer network

Packet routing ✓

Connecting different networks ✓

Traffic management and Security ✓

Network address translation. ✓

OF

7. Differentiate between PAN and MAN

PAN personal area network. is a ~~very~~ small network designed for only personal use and for short range while MAN metropolitan area network covers a metropolitan area and connects multiple LANs in a city.

8. Most suitable Network type for training institution

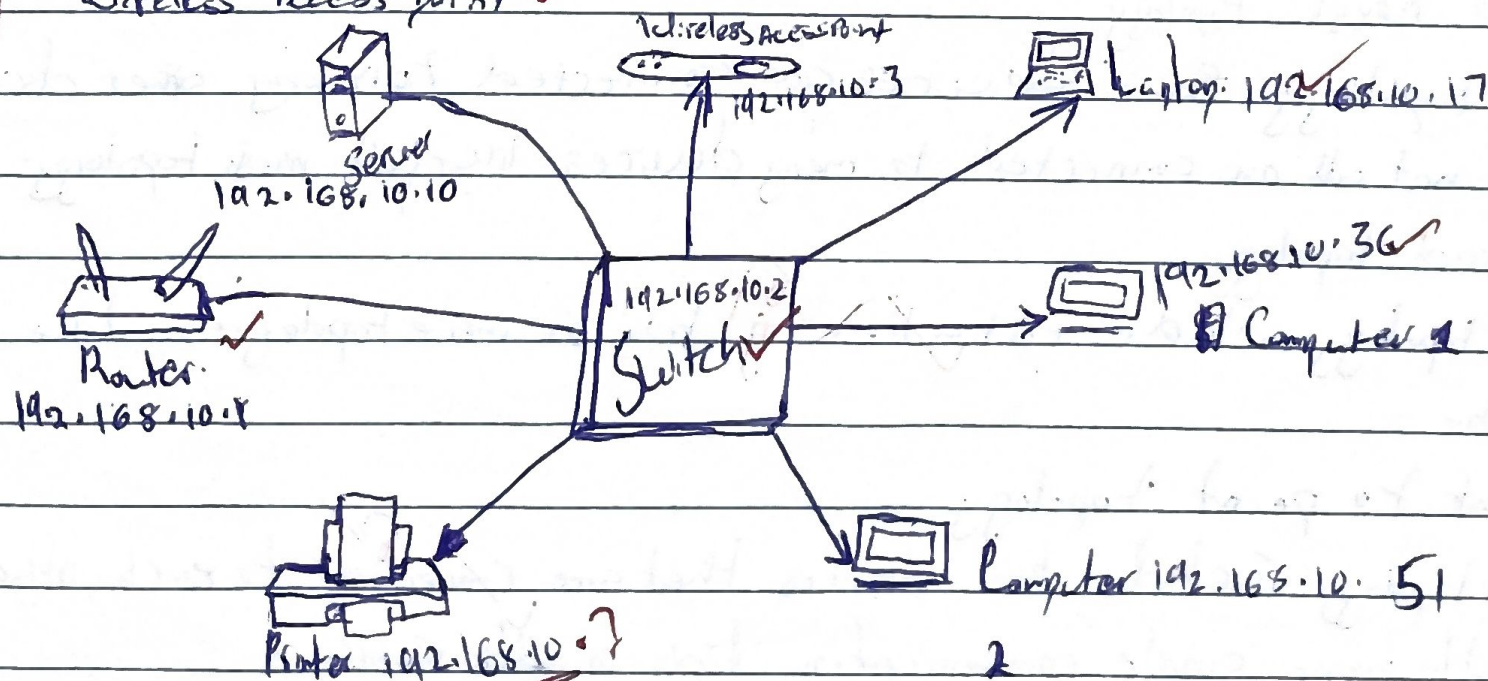
LAN is the most suitable Coz. The organization typically operates within a single building.

Essential devices for LAN setup

Router ✓

Switch ✓

9. Wireless access point ✓



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